Kehelwala D. G. Maduranga, Ph.D.

Applied Machine Learning Scientist — Optimization & AI Systems

Cookeville, TN — Open to Relocate

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Professional Summary

Applied machine learning scientist with deep expertise in optimization, statistics, and deep learning. Proven experience designing scalable ML systems, improving training efficiency, and developing data-driven models for time series, spatio-temporal forecasting, and financial applications. Strong background in PyTorch-based research and production-oriented experimentation.

Core Skills

- **Machine Learning:** Deep Learning, LLM Architectures, RNNs, CNNs, GNNs, GANs, Reinforcement Learning
- **Optimization:** Gradient-based methods, curvature-adaptive learning rates, hyperparameter optimization
- **Programming:** Python, R, C/C++, SQL
- **Frameworks:** PyTorch, TensorFlow, Hugging Face
- **Data & Systems:** Large-scale experimentation, model evaluation, reproducibility, HPC/GPU environments

Industry & Research Experience

Assistant Professor / Applied ML Researcher 2022 – 2026
Tennessee Technological University

- Designed and implemented deep learning models for spatio-temporal forecasting, financial time series, and environmental data.
- Developed optimization techniques improving training stability and computational efficiency for neural networks.
- Led applied ML projects integrating statistical modeling with deep learning architectures.
- Mentored and managed research projects resulting in peer-reviewed publications and deployed prototypes.

Postdoctoral Research Associate (Machine Learning) 2020 – 2022
University of Notre Dame

- Conducted research on hyperparameter optimization and surrogate modeling for deep learning systems.
- Built scalable experimentation pipelines using PyTorch and HPC resources.
- Collaborated with interdisciplinary teams on computer vision and optimization-focused ML research.

Selected Projects & Impact

- **KriGNN: Hybrid Kriging + Graph Neural Networks** — Spatio-temporal air pollution forecasting with improved accuracy and uncertainty modeling (IEEE ICDM Workshop).
- **Curvature-Adaptive Learning Rate Optimizer** — Novel optimizer improving convergence stability and training efficiency (FLAIRS-38).

- **Cryptocurrency Time-Series Forecasting** — RNN-based models for volatile financial markets, focusing on real-time prediction and strategy optimization.
- **HomOpt** — Homotopy-based hyperparameter optimization framework improving search efficiency for deep models.

Education

Ph.D. in Mathematics 2020
University of Kentucky
Focus: Optimization and Structure in Deep Neural Networks

M.S. in Mathematics 2017
University of Kentucky

B.S. (Honors) in Mathematics 2009
University of Sri Jayewardenepura
Minor: Computer Science, Physics

Publications (Selected)

- **HomOpt: A Homotopy-Based Hyperparameter Optimization Method**, Neural Computing and Applications, 2025.
- **Curvature-Adaptive Learning Rate Optimizer**, FLAIRS-38, 2025.
- **Complex Unitary Recurrent Neural Networks**, AAAI, 2019.

Additional

- **Languages:** English (fluent), Sinhala (fluent)
- **Work Authorization:** U.S. employment authorization pending

References

Dr. Qiang Ye, Professor, University of Kentucky
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